

Top Power BI Interview Questions with Answers

1 What is Power BI?

- **Power BI is a business analytics tool by Microsoft that allows users to visualize data, build interactive reports, and gain insights. It connects to various data sources, transforms raw data, and helps in decision-making through real-time dashboards and reports.**

2 Difference between Power BI Desktop and Power BI Service

- **Power BI Desktop: A Windows application used for building and designing reports. It allows data import, modeling, and creating visuals.**
- **Power BI Service: A cloud-based platform where reports/dashboards can be published, shared, and accessed online. It supports collaboration, scheduled refresh, and sharing.**

3 What are the main components of Power BI?

- **Power BI Desktop – For report development**
- **Power BI Service – Online sharing and collaboration**
- **Power BI Mobile – Mobile access to reports**
- **Power Query – Data transformation**
- **Power Pivot – Data modeling**
- **Power View – Data visualization**
- **Power Map – Geo-spatial visualization**

4 What is Power Query?

- **Power Query is a data connection and transformation tool used in Power BI to import, clean, and reshape data from various sources before loading it into the model. It uses the M language and provides a no-code editor for ETL operations.**

5 What is DAX in Power BI?

- **DAX (Data Analysis Expressions)** is a formula language used in Power BI for creating custom calculations, measures, and calculated columns. It is essential for data modeling and creating business logic inside the report.

6 What are measures and calculated columns?

- **Measures:** Calculations evaluated at query time (e.g., SUM of sales). Created using DAX. Used in visuals.
- **Calculated Columns:** New data columns added to tables. Evaluated during data load or refresh. Used for data transformation.

7 How do you create relationships between tables?

- **Relationships** are created in the Model view by dragging one column (usually a key) to a matching column in another table. You can also define them manually by selecting:
 - **Primary & foreign keys**
 - **Cardinality (One-to-One, One-to-Many)**
 - **Cross-filter direction (Single/Both)**

8 What is a data model?

- **A data model is the structured combination of tables, relationships, and business logic (DAX measures, calculated columns) within Power BI that defines how data is stored, connected, and queried for reporting.**

9 What are slicers in Power BI?

- **Slicers are visual filters in Power BI reports that allow users to select values and dynamically filter other visuals. They provide an intuitive way to control which data is displayed (e.g., filter by year or region).**

10 Difference between slicers and filters?

- **Slicers: Visual elements placed on reports; end-users can interact with them. Visible on the report page.**
- **Filters: Applied in the filter pane (visual-level, page-level, report-level); not always visible to users.**